

Intro Khovanov Homology

GST 11/19

Recall. ~~Knot~~ ^{link} invariants are like functions:

$$f: \{\text{link diagrams}\} \longrightarrow A = \{\text{^{mathematical} algebraic objects}\}$$

$$L_1 = L_2 \implies f(L_1) = f(L_2) \quad [\text{equal in some sense, maybe } \cong]$$

Motivation.

- Khov. homol. should be a better invariant
- Khov. ~~original~~ (co)homology theory took link diagrams to bigraded $\mathbb{Z}$ -modules (1999)
- Thrm (Bar-Natan, 2005). Khov. homol. is a stronger invariant than the Jones polyⁿ.
- Rmk. In 2004, Rasmussen used Kh to give a strictly combinatorial proof to the Milnor conj: Slice genus of a (p,q) -torus knot is $\frac{(p-1)(q-1)}{2}$.
- Thrm (Kronheimer and ~~the~~ Mrowka, 2010). Khovanov homology detects the unknot.

→ Recall. It is still an open problem whether or ~~not~~ the Jones polynomial can do this.

this is what has come to be known as a link homology theory (really a cohomology theory)

- In 2006, Khov. classified all Frobenius systems of rank 2 that give rise to link homology theories.

• In (2006), Khovanov classified all Frobenius systems of rank 2 that give rise to link homology theories via his original construction.

• We will work with the system corresponding to the Frobenius algebra $\mathbb{Z}[i, a, h][X]/(X^2 - hX - a)$
 $A = \mathbb{Z}[i, a, h][X]/(X^2 - hX - a)$
 Frobenius algebra w/ a bilinear form mult. and comult maps w/ some properties

Recall Category Theory:

Def. A category C consists of a collection of objects, denoted $Ob(C)$ and $\forall X, Y \in Ob(C)$ a collection of morphisms $f: X \rightarrow Y$ denoted $Hom_C(X, Y)$. Composition should be well defined and there should be an identity map in $Hom_C(X, X)$ $\forall X \in Ob(C)$.

A category is preadditive if $Hom_C(X, Y)$ is an abelian group $\forall X, Y \in Ob(C)$.

Ex. ~~$\mathbb{G}_p$~~ Grp and $R\text{-mod}$ are pre-additive

Given $f: G \rightarrow H$ and $g: G \rightarrow H$, $f+g: G \rightarrow H$ defined pointwise: ~~$(f+g)(g) = f(g) + g(g)$~~

$$(f+g)(a) = f(a) + g(a) \text{ and } (f+g)(a+b) = f(a+b) + g(a+b) \\ = f(a) + f(b) + g(a) + g(b) \\ = \cancel{f(a)} + f(g(a)) + (f+g)(b). \checkmark$$

Functors. Functors are maps between categories. They send objects to objects and morphisms to morphisms.

Let $X, Y \in Ob(C)$, $F: C \rightarrow D$ a functor, $f \in Hom_C(X, Y)$

If $F(f): F(X) \rightarrow F(Y)$, F is covariant, if

$F(f): F(X) \leftarrow F(Y)$, F is contravariant.

Ex. $\Pi_n : \text{Top}_* \rightarrow \text{Grp}$ $\Pi_n(X, x_0)$ is the object in grp, if $\varphi : (X, x_0) \rightarrow (Y, y_0)$, $\varphi_* : \Pi_n(X, x_0) \rightarrow \Pi_n(Y, y_0)$ is a gp. hom.

$\text{Spec} : \text{Rings} \rightarrow \text{Top}$ $R \mapsto \text{Spec}(R)$
 $\underbrace{\text{set of prime ideals w/ Zariski topology}}$

$f : R \rightarrow S$ induces a map $f_* : \text{Spec}(S) \rightarrow \text{Spec}(R)$.

The Category of Interest.

- Given a preadditive category C , we can construct new categories

- $\text{Mat}(C)$ and $\text{Kom}(C)$.

$\text{Mat}(C)$ consists of objects $\bigoplus_{i=1}^n X_i$ where $X_i \in \text{Ob}(C)$
 morphisms in $\text{Mat}(C)$ are matrices. if $F \in \text{Hom}_{\text{Mat}(C)}(X, Y)$
 where $X = \bigoplus_{i=1}^m X_i$ and $Y = \bigoplus_{j=1}^n Y_j$ then $F = (f_{ij})$ is
 an $m \times n$ matrix where $f_{ij} \in \text{Hom}_C(X_j, Y_i)$

$\text{Mat}(C)$ is preadditive, add morphisms w/ matrix addition, composition ~~replaces multiplication~~ in $\text{Mat}(C)$ is matrix multiplication, but composition replaces the usual multiplication of entries.

- $\text{Kom}(C)$ is the category of complexes over C , objects are cochain complexes of finite length

$$0 \rightarrow \dots \rightarrow X^{r-1} \xrightarrow{d^{r-1}} X^r \xrightarrow{d^r} X^{r+1} \xrightarrow{d^{r+1}} \dots \rightarrow 0$$

i.e. $d^r \circ d^{r-1} = 0 \quad \forall r$, where $X^r \in \text{Ob}(C) \quad \forall r$ and $d^r \in \text{Hom}_C(X^r, X^{r+1}) \quad \forall r$. Morphisms are chain maps that yield commutative diagrams

$$\begin{array}{ccccccc} 0 & \rightarrow & \dots & \rightarrow & X^{r-1} & \xrightarrow{d^{r-1}} & X^r & \xrightarrow{d^r} & X^{r+1} & \rightarrow & \dots & \rightarrow & 0 \\ & & & & \downarrow & & \downarrow & & \downarrow & & & & \downarrow \\ \phi & \rightarrow & \dots & \rightarrow & Y^{r-1} & \xrightarrow{d^{r-1}} & Y^r & \xrightarrow{d^r} & Y^{r+1} & \rightarrow & \dots & \rightarrow & 0 \end{array}$$

Webs

Def. A web Γ with boundary B is a 1-manifold embedded in a disk D^2 , and decomposed into oriented arcs. At each vertex arcs are oriented

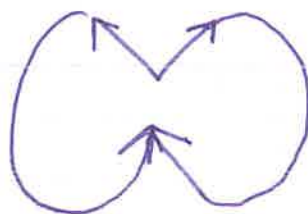
inwards  or outwards 

We allow webs w/o boundary as well (closed webs). (planar graphs w/ bivalent vertices, and circles).

E.g.



$$|B| = 6$$



Closed web

- we want a category whose objects are webs w/ boundary B

Foams

Def. Let Γ_0 and Γ_1 be webs w/ boundary B .

{ A foam w/ boundary B is an abstract cobordism from Γ_0 to Γ_1 regarded up to boundary preserving isotopy.

That is, a foam is a piecewise oriented 2-manifold S with boundary $-\Gamma_1 \cup \Gamma_0 \cup (B \times [0, 1])$ and corners where $-\Gamma_1$ denotes Γ_1 with opposite orientation.

Rmk. - Webs contained in disks $\Rightarrow$ foams contained

- Foams between closed webs are just ^{in cylinders} ~~just~~ contained in $\mathbb{R}^2 \times [0, 1]$ and boundary

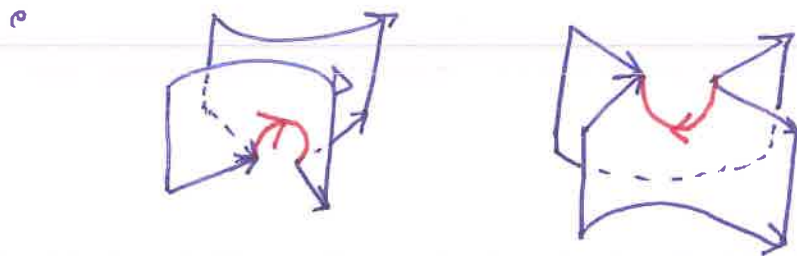
$\longleftarrow \mathbb{R}^2 \times \{0, 1\}$.

- Foams may contain singular arcs or circles where the orientations of the webs disagree

- Singular arcs separate facets (2-cells in the CW-complex structure)

- CW complex structure only important for computing $X(s)$ for the grading.

Ex. $\circ$  $= \text{id}(\rightarrow)$

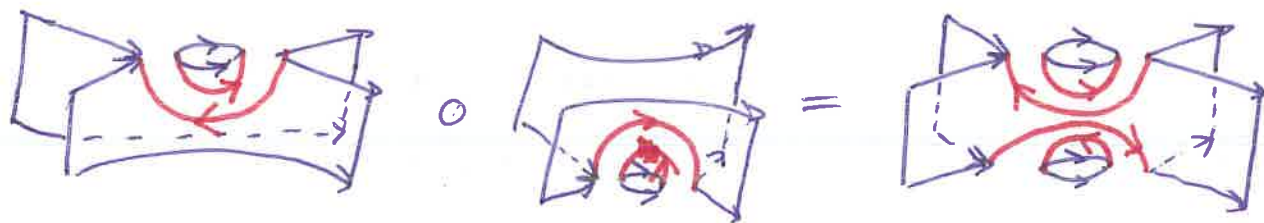


- Foams are read from bottom to top by convention.

- we allow dots to occupy the facets of the foams.
- allow $\mathbb{Z}[i, a, h]$ -linear combos of dotted foams

Def $\text{Foams}(B)$ is the category of ~~dotted~~ webs ~~foams~~ and dotted foams between them

$i = \sqrt{-1}$, a, h are parameters. With $\mathbb{Z}[i, a, h]$ -lin combos, $\text{Foams}(B)$ (or $\text{Foams}(\emptyset)$) the $\text{Hom}(-, -)$ sets inherit an abelian gp structure. It is a preadditive category where composition is done by stacking morphisms on top of one another.



We consider the local relations below and define $\text{Foams}_{\mathbb{Z}[h]}$ the category of Foams ~~mod~~ where 2 morphisms are identified if they differ by a finite series of local relations here:

(2D)  = h  + a 

(SF)  =  +  - h 

(S)  = 0

(UFO)  = 0 = ,  = i = - 

- The motivation in taking the quotient modulo these relations is in ~~the~~ showing our chain complex is an invariant at the geometric level. and the construction of a functor ~~such a~~
 $F: \text{Foams}_{\mathbb{Z}[h]} \rightarrow \mathbb{Z}[i, h] - \text{Mod}.$

Such a functor will extend immediately to a functor

$$F: \text{Mat}(\text{Foams}_{\mathbb{Z}[h]}) \rightarrow \text{Mat}(\mathbb{Z}[i, h] - \text{Mod})$$

and again to

$$F: \underbrace{\text{Kom}_{\mathbb{Z}[h]}(\text{Mat}(\text{Foams}_{\mathbb{Z}[h]}))}_{:= \text{Kof}_{\mathbb{Z}[h]}} \rightarrow \text{Kom}_{\mathbb{Z}[h]}(\text{Mat}(\mathbb{Z}[i, h] - \text{Mod})) \quad \textcircled{7}$$

Now $\text{Foams}(B)$ is graded by degree:

Given a foam S with d dots we

define $\deg(S) = -\chi(S) + 2d + \frac{1}{2}|B|$,

$\chi(S)$ is the Euler char.

If time permits:

Def. ~~A~~ A pre additive category C is graded if:

- $\forall X, Y \in \text{Ob}(C)$, $\text{Hom}_C(X, Y)$ is a graded abelian grp s.t.
 $\deg(g \circ f) = \deg g + \deg f$ and
 $\deg(\text{id}_X) = 0 \quad \forall \text{id}_X \in \text{Hom}_X(X, X)$.
- $\mathbb{Z}$ -action $(m, X) \mapsto X \{m\}$ called the grading shift on objects. It doesn't change objects, but
 $\text{Hom}(X, Y) = \text{Hom}(X \{m_1\}, Y \{m_2\})$ as sets
while if $f \in \text{Hom}(X, Y)$ has degree d ,
then $\deg(f) = d + m_2 - m_1$, when
 $f \in \text{Hom}(X \{m_1\}, Y \{m_2\})$.

We impose the grading

$\deg(1) = 0 = \deg(i)$, $\deg(h) = 2$, $\deg(a) = 4$,

(then the relations in $\mathcal{L}$ respect the degree
on the ring $\mathbb{Z}[i, a, h]$

⑧

SL(2) polynomial

- To each crossing  or  in a link diagram, ~~as~~ replace w/ a smoothing  or . The result is a web Γ w/ empty boundary. We evaluate webs Γ locally via

$$(1) \langle \bigcirc \cup \Gamma \rangle = (q + q^{-1}) \langle \Gamma \rangle = \langle \bigcirc \cup \Gamma \rangle$$

$$(2) \langle \text{crossing with arrows} \rangle = \langle \text{smoothing} \rangle, \langle \text{crossing with arrows} \rangle = \langle \text{smoothing} \rangle$$

Now impose the rule

$$(3) \langle \text{crossing with arrows} \rangle = q \langle \text{smoothing} \rangle - q^2 \langle \text{crossing with arrows} \rangle$$

$$(4) \langle \text{crossing with arrows} \rangle = q^{-1} \langle \text{smoothing} \rangle - q^{-2} \langle \text{crossing with arrows} \rangle$$

Then ~~define~~ given a link diagram ~~D~~ D ,
 $P_L(q) =$ the bracket polynomial is
 defined

$$\langle D \rangle = \sum_{\Gamma} \pm q^{\alpha} \langle \Gamma \rangle \in \mathbb{Z}[q, q^{-1}]$$

is a link invariant.

Proof omitted.

$P_L(q) := \langle D \rangle$ is the quantum
 SL(2) link invariant.

The geometric complex

$$\begin{aligned} \bullet [\text{crossing}] &= 0 \rightarrow \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{23\} \xrightarrow{-1} \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{13\} \rightarrow 0 \\ \bullet [\text{crossing}] &= 0 \rightarrow \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{-13\} \xrightarrow{1} \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{-23\} \rightarrow 0 \end{aligned}$$

- $[T]$ is a formal tensor product of the complexes associated to each crossing of T .
- The complex $[T]$ is nontrivial in the chain spaces $-n_+$ through n_- .

Claims

- $[T] \in \text{Kof}_n(B)$
i.e. $d^r \circ d^{r-1} = 0$ & differentials.
- $[T]$ is a tangle invariant
i.e. $T \underset{\text{isotopy}}{\sim} T' \Rightarrow [T] \underset{\text{homotopy equivalent}}{\simeq} [T']$

Saddles can be time reordered in a cobordism via isotopy

• In words: ~~associate~~ number all crossings of T construct an n -dim. hypercube, labeling vertices as binary strings associated to what state of what crossing appears. To the edges of the hypercube associate a foam between the states.

Some negative signs appear

$$\begin{array}{ccccc} \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{23\} & \xleftarrow{0} & \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} & \xrightarrow{1} & \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{13\} \\ \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{-13\} & \xleftarrow{0} & \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} & \xrightarrow{1} & \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \{-23\} \end{array}$$

- Show Bar-Natan's trefoil
 - Show invariance of complex for $R1$ and $R2$
- slides

Functor $F_\emptyset$

- Recall $\mathbb{Z}[i, a, h]$ is a graded ring w/ $\deg(1) = 0 = \deg(i)$, $\deg(a) = 4$, $\deg(h) = 2$.

$A = \mathbb{Z}[i, a, h][X] / (X^2 - hX - a)$ is a graded $\mathbb{Z}[i, a, h]$ -module with

$\deg(1) = -1$ and $\deg(X) = 1$.
of rank 2.

really, A is a $\mathbb{Z}[i, a, h]$ -algebra under the inclusion map $i: \mathbb{Z}[i, a, h] \hookrightarrow A$.

- Note $\text{Hom}_{\text{Foams}/\mathbb{Z}}(\emptyset, \Gamma)$ is an abelian group, and $\mathbb{Z}[i, a, h]$ acts on its elements in the obvious way, so $\text{Hom}_{\text{Foams}/\mathbb{Z}}(\emptyset, \Gamma)$ is a $\mathbb{Z}[i, a, h]$ -module.

Def. $F_\emptyset : \text{Foams}_h(\emptyset) \rightarrow \mathbb{Z}[i, a, h]\text{-Mod}$

- $F_\emptyset(\Gamma) = \text{Hom}_{\text{Foams}_h(\emptyset)}(\emptyset, \Gamma) \quad \forall \Gamma \in \text{Ob}(\text{Foams}_h(\emptyset))$
- $\forall S \in \text{Hom}_{\text{Foams}_h(\emptyset)}(\Gamma', \Gamma''), F_\emptyset(S)$ is a $\mathbb{Z}[i, a, h]$ -linear map $\text{Hom}_{\text{Foams}_h(\emptyset)}(\emptyset, \Gamma') \rightarrow \text{Hom}_{\text{Foams}_h(\emptyset)}(\emptyset, \Gamma'')$
 $u \mapsto S \circ u$.

Rmk. The grading on $\deg(s) = -X(s) + 2d + \frac{1}{2}|B|$ induces a natural grading on the ~~to~~ objects $F_\emptyset(\Gamma)$, so $F_\emptyset$ ~~respects~~ is degree-preserving by construction

• as mentioned before, $F_\emptyset$ extends to a functor $K\text{of}_h = \text{Kom}_h(\text{Mat}(\text{Foams}_h(\emptyset))) \rightarrow \text{Kom}(\text{Mat}(\mathbb{Z}[i, a, h]\text{-Mod}))$

• dots on the foam represent multiplication by X in the algebra A

$$X^2 = hX + a \quad \text{and} \quad (ED) \text{ rel for e.g.}$$

Prop. The functor F_ϕ mimics the web s.r.'s for in the sense that we have the following isomorphisms of abelian groups:

$$(1) F_\phi(\bigcirc) \cong A \cong F_\phi(\bigcirc)$$

$$(2) F_\phi(\Gamma \cup \bigcirc) \cong F_\phi(\Gamma) \otimes_{\mathbb{Z}[i, q, h]} A \cong F_\phi(\Gamma \cup \bigcirc)$$

$$(3) F_\phi(\text{crossing}) \cong F_\phi(\text{cup}), F_\phi(\text{cap}) \cong F_\phi(\text{cup})$$

Remark Since k homotopy equiv. complexes have isomorphic (co)homology, $H(F_\phi[L])$ is an invariant (up to homotopy) for the link L , with $\mathcal{H}^{i,j}(L) = H^{i,j}(F_\phi(L))$, i is the homology ~~grp~~, j is the quantum grading.

~~j is the~~ $\mathcal{H}^{i,j}(L)$ is the j^{th} abelian gp in the direct sum decomp of the graded ~~module~~, $\mathbb{Q}$ ~~grp~~ ab group in the i^{th} cohomology group of the chain complex $F_\phi[L]$

Since $F_\emptyset$ respects web s.r., by construction

$$\begin{aligned} P_L(q) &= \sum_{i,j \in \mathbb{Z}} (-1)^i q^j \text{rank } \mathcal{H}^{i,j}(L) \\ &= \sum_{i,j \in \mathbb{Z}} (-1)^i q^j \text{rank } H^{i,j}(F_\emptyset[L]). \end{aligned}$$

Rmk. When $a=0=h$, this is isomorphic to Khovanov's og. homology theory.

- The i in the ground ring and use of sealed cobordisms helped to fix the sign ambiguity in the functorial property.

Rmk. $H^i(L) = \bigoplus_{j \in \mathbb{Z}} H^{i,j}(L)$